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RESEARCH ARTICLE

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Abstract

The ICH E17 guideline describes regional pooling in multi-regional clinical trials (MRCTs) in terms of the similarity of effect modifier distributions but specifies no similarity measure, and, to our knowledge, no distance between regional effect modifier distributions has been evaluated for this purpose. We quantify how two regions' distributions of each candidate effect modifier differ by the Wasserstein-1 (W_1) distance, with bootstrap confidence intervals. If the conditional average treatment effect is common to all regions, depends on that effect modifier alone, and is Lipschitz with clinical slope L_{clinical}, regional average treatment effects differ by at most $\Delta_{\max} = L_{\text{clinical}} W_1$; when L_{clinical} is unknown, the smallest slope L^* at which the distance could produce a pre-specified clinical margin is compared with a sponsor-specified bound. In simulations, the standardized mean difference remained at chance in some configurations up to 2,000 patients per region, whereas W_1 and the Kolmogorov–Smirnov statistic reached the discrimination target in every configuration, W_1 with fewer patients in all but a uniform-shift control. For all but the smallest non-null difference, bootstrap coverage was 0.92–0.95 from 100 patients per region. In a hypothetical GUSTO-I-based MRCT, partner rankings on age and systolic blood pressure differed. The framework quantifies differences; the pooling judgment remains clinical.

KEYWORDS
multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

1 | INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become the standard paradigm for global pharmaceutical development.[1,2] The International Council for Harmonisation (ICH) E17 guideline, adopted in 2017, established principles for planning and designing MRCTs to evaluate whether treatment effects can be regarded as generalizable across the target population.[3]

Regulatory authorities in Japan and China expect demonstration of treatment effect consistency in regional subpopulations,[4,5] yet individual regional sample sizes in global MRCTs are often insufficient for such demonstration. A key strategy for addressing this challenge is the regional pooling approach, wherein regions with similar patient characteristics are grouped for analysis to enhance the ability to assess regional consistency.[3] The ICH E17 guideline describes pooling decisions in terms of the similarity of effect modifier distributions.[3]

An effect modifier is a baseline patient characteristic—such as age, disease severity, or genetic marker—for which the treatment benefit differs across subgroups. When such heterogeneity exists, even if the drug works identically at the individual level, regions with different patient compositions may observe different average treatment effects. A region with predominantly

Abbreviations: AMI, acute myocardial infarction; AUC, area under the receiver operating characteristic curve; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; CRP, C-reactive protein; CV, coefficient of variation; EDF, empirical distribution function; GUSTO, Global Utilization of Streptokinase and t-PA for Occluded coronary arteries; ICH, International Council for Harmonisation; IPD, individual patient data; KL, Kullback–Leibler; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; MRI, magnetic resonance imaging; NMPA, National Medical Products Administration; RMSE, root mean squared error; RWE, real-world evidence; SBP, systolic blood pressure; SD, standard deviation; SMD, standardized mean difference.

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younger patients would show larger benefits than a region with predominantly older patients, not because the drug works differently, but because the patient mix differs. This fundamental relationship underscores why the similarity of the full effect modifier distribution, not only its average level, is critical to the validity of regional pooling. In this paper, we focus on continuous effect modifiers, such as age and disease severity scores; categorical effect modifiers raise distinct methodological considerations and are outside the present scope.

The extent to which the relevant effect modifiers are known varies across development programmes, and this shapes what can be planned quantitatively.[6] For some treatments, mechanism or prior trials identify a specific effect modifier in advance—a genetic marker known to determine response, for instance—so the characteristic requiring cross-regional comparison is established before planning. More often at the planning stage, the effect modifiers are only partially understood, and the sponsor proceeds from candidate characteristics suggested by biological plausibility or earlier-phase data, or from broader regional and ethnic factors that stand in as surrogates for the underlying modifiers.[6] The framework developed here operates on whichever set of candidate effect modifiers the sponsor specifies: it quantifies the cross-regional distributional similarity of each candidate, orders regions by it, and calibrates it clinically, and is agnostic to how the candidates were identified.

Despite the regulatory importance of effect modifier distributional similarity, current practice lacks a standardized, clinically calibrated methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are "similar enough." Song et al., writing on the NMPA's implementation of ICH E17, identify the pooling strategy as an area where ambiguity remains and describe clustering populations on distances defined from the correlation between effect modifiers and endpoints, citing the reference volume discussed next.[5] The pooling-strategy chapter of a reference volume on multi-regional trials under ICH E17 introduces, as one of many possible approaches, cluster analysis of regions on region-level summaries of the candidate effect modifiers: each region is given a representative value for each candidate effect modifier, the values are typically standardized, and distances such as the Euclidean distance between regions are weighted for relevance, for instance through a Lasso regression on the outcome; the chapter notes that the selection of the distance measure is one of the key decisions, and does not evaluate the approach.[6] Two gaps remain. First, to our knowledge no distance between regional effect modifier distributions has been proposed and evaluated for this purpose: a single summary per modifier leaves two regions that agree on it but differ in spread, shape, or tail behaviour at distance zero—for a binary characteristic, as in the chapter's illustration, one proportion describes the distribution completely, so the limitation concerns continuous modifiers—and any finite set of summaries requires deciding in advance which distributional features are relevant, rather than comparing the full distributions directly. Second, the chapter calls for a judgment of whether differences in the impact of effect modifiers across pooled regions are "clinically meaningful or not" but supplies no metric to operationalize that judgment.

These gaps bear on the planning stage, when the pooling strategy for small-sample regions is specified. A workshop report by Matsushima et al. documented that regional imbalance in CRP-positive/MRI-negative status contributed to apparent inconsistency in the secukinumab MRCT.[4] This example illustrates the practical consequences when effect modifier distributional differences are not assessed quantitatively at the planning stage. That case involved a binary status, where imbalance reduces to a difference in proportions; the continuous effect modifiers on which we focus can differ in shape and spread, not only average level, so assessing their similarity requires more than comparing means.

General-purpose comparison tools such as the standardized mean difference are routinely applied to baseline covariates,[7] but the standardized mean difference captures only differences in location: regions can match on means yet differ in variance or distributional shape. When effect modification is non-linear, these features directly influence regional average treatment effects.

This paper addresses these gaps by proposing a quantitative framework for assessing effect modifier distributional similarity across regions, based on the Wasserstein-1 (W_1) distance between two cumulative distribution functions, retained in the original measurement units of the effect modifier. Geometrically, W_1 equals the area between the two cumulative distribution functions. Unlike the standardized mean difference, W_1 responds to differences in variance and shape as well as location. We estimate W_1 with bootstrap confidence intervals and link it to clinical interpretation through a calibration framework.

The remainder of this paper is organized as follows. Section 2 presents the methodological framework: existing measures and their limitations, the definition of the per-effect-modifier W_1 distance, the estimation procedure, and its clinical calibration. Section 3 presents two simulation studies, evaluating identification and estimation properties. Section 4 illustrates the framework with publicly available individual patient data from the GUSTO-I trial. Section 5 discusses implications, limitations, and future directions.

2 | METHODS

An effect modifier is a factor for which the treatment effect varies across subgroups. Formally, let the conditional average treatment effect (CATE) be denoted $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, where X is the effect modifier. When this function is non-constant, the average treatment effect observed in region r depends on the distribution F_r of the effect modifier in that region:

$$\bar{\tau}_r = \int \tau(x)\, dF_r(x). \tag{1}$$

Assessing pooling suitability therefore requires a measure of distributional difference between regions.

2.1 | Existing Approaches and Their Limitations

The most widely used measure for comparing covariate distributions across groups is the standardized mean difference (SMD), defined as

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_2}{S_{\text{pooled}}}, \tag{2}$$

where $\bar{X}_r$ denotes the sample mean in region r and S_{pooled} is the pooled standard deviation.[7] Because treatment effects in clinical trials are typically summarized as mean differences, SMD provides a scale-free comparison directly relevant to the quantities that drive clinical decision-making. However, by construction SMD captures only differences in location. Two distributions with identical means but markedly different variances—for example, $N(50, 5^2)$ and $N(50, 15^2)$—yield SMD = 0 despite a threefold difference in spread.

Distributional distance measures address this limitation by comparing distributions beyond summary statistics. Empirical distribution function (EDF) tests such as the Kolmogorov–Smirnov (KS) statistic, $D_{\text{KS}} = \sup_x |F_1(x) - F_2(x)|$, and related measures (Cramér–von Mises, Anderson–Darling) quantify differences in the full shape of distributions by comparing CDFs directly. Information-theoretic divergences such as Kullback–Leibler (KL) divergence,

$$D_{\text{KL}}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx, \tag{3}$$

take a density-based approach, measuring how one probability distribution differs from another.

However, each of these approaches has limitations when applied to comparing effect modifier distributions across regions. SMD captures only location differences and is blind to variance and shape. EDF-based statistics such as the KS statistic capture full distributional differences but lack a clinically interpretable scale, and the bound on treatment effect heterogeneity they admit carries a constant that clinical sources do not report (Section 2.2). KL divergence is asymmetric—comparing region A to region B yields a different value than comparing B to A—can diverge to infinity when empirical distributions do not fully overlap, and requires density estimation that is impractical with small regional samples. None of these measures offers a theoretical link that would translate a distributional distance into a bound on regional treatment effect differences whose constant is a per-unit clinical slope, the kind of quantity that treatment-by-covariate interaction analyses estimate. These gaps motivate a measure that (i) captures distributional features beyond location, (ii) provides a clinically interpretable scale, and (iii) offers such a link. The Wasserstein-1 distance and its clinical calibration, developed in the following subsections, satisfy all three requirements.

2.2 | Per-Effect-Modifier Wasserstein-1 Distance

Among distributional distances that could capture the full distributional differences relevant to regional pooling, we adopt the Wasserstein-1 distance (W_1) for the clinical specifiability of the constant in its heterogeneity bound (Proposition 1 below). The W_1 distance (sometimes called the Earth Mover's Distance) admits three equivalent representations on the real line, each highlighting a different facet of its interpretation.[8,9]

Let F_1 and F_2 denote the cumulative distribution functions (CDFs) of an effect modifier in two regions. Then

$$W_1(F_1, F_2) = \int_{-\infty}^{\infty} |F_1(x) - F_2(x)|\, dx = \int_0^1 |F_1^{-1}(u) - F_2^{-1}(u)|\, du = \inf_{\pi \in \Pi(F_1, F_2)} \int |x - y|\, d\pi(x, y), \tag{4}$$

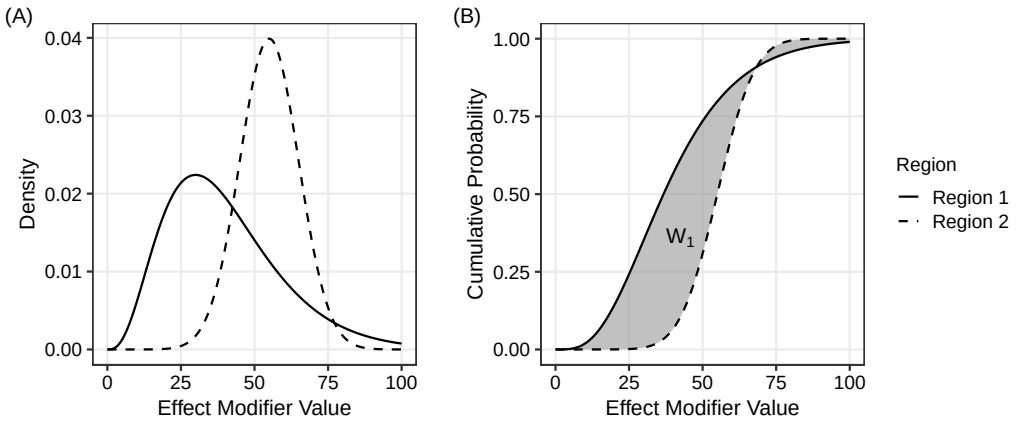


FIGURE 1 The Wasserstein-1 distance as the area between cumulative distribution functions, illustrated with a right-skewed Region 1 (gamma distribution) and a symmetric Region 2 (normal distribution). Panel (A): probability densities. Panel (B): cumulative distribution functions; the shaded area equals $W_1(F_1, F_2)$.

where F_r^{-1} is the quantile function and $\Pi(F_1, F_2)$ is the set of probability couplings with marginals F_1 and F_2. The CDF form expresses W_1 as the area between the two CDFs (Figure 1); the quantile form as the integrated absolute difference of quantile functions; and the Kantorovich form as the minimum expected absolute displacement to transport one distribution onto the other. The three forms are mathematically equivalent.[9] Unlike the standardized mean difference, W_1 responds to differences in variance, skewness, and other distributional features.[10]

A property of W_1 central to our framework is that it carries the units of the underlying effect modifier. In each of the three equivalent forms in equation (4), W_1 carries the units of X: the CDF form integrates a dimensionless difference of probabilities over x, the quantile form integrates differences of quantiles, and the coupling form integrates distances $|x - y|$. Consequently, $W_1(F_1, F_2)$ is expressed on the original scale of the effect modifier. This dimensional alignment supports direct clinical interpretation: a regional $W_1 = 5$ mmHg in systolic blood pressure means that optimally transporting one distribution onto the other moves probability mass an average distance of 5 mmHg. The same logic precludes a meaningful direct comparison of W_1 values across effect modifiers measured on different scales, and we therefore restrict attention in this paper to per-effect-modifier assessment (Section 5 elaborates on this scope choice).

W_1 is a proper metric on the space of probability distributions with finite first moment, satisfying non-negativity, symmetry, identity of indiscernibles, and the triangle inequality.[8] It is also affine-equivariant in the variable units, satisfying $W_1(F_{aX+b}, F_{aY+b}) = |a| \cdot W_1(F_X, F_Y)$ for any $a \neq 0$. Beyond these basic properties, W_1 admits a Kantorovich–Rubinstein dual representation[8]

$$W_1(F_1, F_2) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left| \int f \, dF_1 - \int f \, dF_2 \right|, \tag{5}$$

which expresses W_1 as the largest possible difference in expected values of a 1-Lipschitz function f between the two distributions (the notation $\|f\|_{\text{Lip}} \leq 1$ means $|f(x) - f(y)| \leq |x - y|$ for all x, y). If the conditional average treatment effect function $\tau(x)$ is Lipschitz continuous with constant L_{clinical}, so that $|\tau(x) - \tau(y)| \leq L_{\text{clinical}}|x - y|$ for all x, y, then τ/L_{clinical} is 1-Lipschitz and applying equation (5) yields the central result of this section:

Proposition 1. (Connection to heterogeneity). *Suppose the CATE function τ is the same in every region and depends on the effect modifier X alone. If τ is Lipschitz continuous with constant $L_{clinical}$, then for any two regions with effect modifier distributions F_1 and F_2,*

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L_{clinical} \cdot W_1(F_1, F_2). \tag{6}$$

Proof. Apply equation (5) with $f = \tau/L_{\text{clinical}}$, which is 1-Lipschitz by assumption. This yields $\left|\int (\tau/L_{\text{clinical}})\, d(F_1 - F_2)\right| \leq W_1(F_1, F_2)$. Combining with equation (1) for $\bar{\tau}_r$ and rearranging yields equation (6). □

Equation (6) is not the only bound of its form. Integration by parts gives $\bar{\tau}_1 - \bar{\tau}_2 = -\int (F_1(x) - F_2(x))\, d\tau(x)$, so for a CATE of bounded variation on a bounded support, $|\bar{\tau}_1 - \bar{\tau}_2| \leq \text{TV}(\tau) \cdot D_{\text{KS}}$ as well, and $W_1 \leq W_2$ makes $L_{\text{clinical}} \cdot W_2$ a valid but looser

bound.[8] What distinguishes equation (6) is its constant. L_{clinical} is a change in treatment effect per unit of the effect modifier, the kind of quantity that treatment-by-covariate interaction analyses estimate; $\text{TV}(\tau)$ is the total excursion of the CATE over the support, which reduces to a clinically stated quantity only for a monotone CATE. Where both constants are available, neither bound dominates in general. On a support $[a, b]$, $W_1 \leq (b-a)\, D_{\text{KS}}$ and $\text{TV}(\tau) \leq L_{\text{clinical}}\,(b-a)$, and $L_{\text{clinical}} \cdot W_1 \leq \text{TV}(\tau) \cdot D_{\text{KS}}$ holds exactly when $W_1/D_{\text{KS}} \leq \text{TV}(\tau)/L_{\text{clinical}}$. For a CATE whose slope has constant magnitude L_{clinical}, such as a linear CATE, $\text{TV}(\tau) = L_{\text{clinical}}(b-a)$ and equation (6) is never the looser of the two; the Kolmogorov–Smirnov bound gains as the CATE's slope falls below its maximum over much of the support, most sharply when the CATE changes steeply over a narrow range—the threshold case in which equation (6) is itself conservative (Section 5).

The right-hand side of equation (6) involves W_1 in its original measurement units, multiplied by a clinical slope L_{clinical}; no further normalization of W_1 is required to obtain a clinically interpretable bound. We therefore adopt W_1 in its original units as the primary similarity metric throughout this paper. One could instead normalize W_1 by a within-effect-modifier scale measure such as the pooled interquartile range; we show in Section 2.4 that such normalization is mathematically redundant once equation (6) is used for clinical calibration. The Lipschitz assumption underlying Proposition 1 is appropriate when the CATE function varies smoothly with the effect modifier; for threshold or step-function CATEs, L_{clinical} can be arbitrarily large and the bound becomes conservative (Section 5).

2.3 | Estimation

Given samples $\{X_{1,i}\}_{i=1}^{n_1}$ from region 1 and $\{X_{2,j}\}_{j=1}^{n_2}$ from region 2, the Wasserstein-1 distance is estimated using empirical distribution functions. Since $\hat{F}_1$ and $\hat{F}_2$ are piecewise constant (right-continuous step functions), the integral in equation (4) reduces to a finite sum over the intervals between consecutive distinct values of the pooled sample:

$$\widehat{W}_1 = \sum_{k=1}^{m-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)}), \tag{7}$$

where $\hat{F}_r$ denotes the empirical CDF (right-continuous) of region r and $x_{(1)} < \cdots < x_{(m)}$ are the m distinct values of the pooled sample (ties are pervasive when an effect modifier is recorded on a coarse grid, as with systolic blood pressure recorded in whole mmHg in Section 4). The estimator $\widehat{W}_1$ inherits the original measurement units of the effect modifier. The asymptotic distribution of W_1 is non-standard: del Barrio et al.[11] showed that $\sqrt{n}\, W_1(\hat{F}_n, F) \xrightarrow{d} \int |B(F(x))|\, dx$, where B is a standard Brownian bridge. Because the quantiles of this limiting distribution depend on the unknown F, no universal critical values are available and direct construction of asymptotic confidence intervals is impractical. We therefore employ the nonparametric percentile bootstrap with $B = 2{,}000$ replicates.[12] In the two-sample setting, when the set $\{x : F_1(x) = F_2(x) \in (0,1)\}$ has Lebesgue measure zero—for example, when the two CDFs cross only at isolated points—the Hadamard derivative of the L_1 functional at $F_1 - F_2$ is linear, and the ordinary nonparametric bootstrap is consistent by the functional delta method.[13] Where the CDFs coincide on an interval, including $F_1 = F_2$, the derivative is non-linear and the naive bootstrap is not consistent.[14,13,15] Convergence occurs at rate $\sqrt{n_1 n_2/(n_1+n_2)}$ (see Appendix A.2 for details). The estimator (7) is computed in $O((n_1+n_2)\log(n_1+n_2))$ time, dominated by sorting the combined order statistics.

Because W_1 is a distance, $\widehat{W}_1$ is non-negative by construction and is typically positive even when the two regions are draws from the same distribution. We refer to the sampling distribution of $\widehat{W}_1$ under $F_1 = F_2$ as the null floor. It is obtained by drawing two independent resamples, of the two regions' actual sizes, from one region's empirical distribution—the anchor's in Section 4.2—and recomputing $\widehat{W}_1$; because it depends on both sample sizes and on the source distribution, it is computed per region pair. An observed $\widehat{W}_1$ at or below its 95th percentile, the reference point used in Section 4.2, is one the estimator would produce with appreciable probability from identical distributions, so it is not evidence of a difference; how large the difference could be is read from the upper confidence limit of $\widehat{W}_1$, equivalently of $\Delta_{\max}$.

2.4 | Interpretation and Clinical Calibration

A central feature of W_1 is its connection to treatment effect heterogeneity through Proposition 1. This connection provides the basis for a clinically grounded interpretation framework rather than reliance on fixed numerical cutoffs.

We refer to the Lipschitz constant of the CATE function $\tau(x)$ as the clinical slope L_{clinical}. By definition, L_{clinical} quantifies the maximum change in the treatment effect per unit change in the effect modifier, expressed on the outcome scale of the trial—for example, absolute 30-day mortality risk per year of age, or per mmHg of systolic blood pressure. Throughout this paper L_{clinical} is quoted on the proportion scale of the outcome, so that $L_{\text{clinical}} = 1 \times 10^{-2}$ per year denotes one percentage point of absolute 30-day mortality per year of age; clinical margins Δ_{clin} stated in percentage points are divided by 100 before entering any calculation. When prior individual patient data (IPD) are available, meta-regression on treatment-by-covariate interaction estimates the average interaction slope.[16,17] An average slope is, in absolute value, a lower bound on the Lipschitz constant rather than the constant itself: for non-linear τ, the local slope can exceed the fitted one. An L_{clinical} specified from such an estimate should therefore carry a sponsor-specified multiplicative safety factor, in the spirit of margin discounting in non-inferiority design, so that the premise of Proposition 1 remains defensible. More commonly at the MRCT planning stage, published subgroup reports document the direction of interaction rather than a per-unit slope, and a precise numerical L_{clinical} is unavailable. We address both situations below.

From Proposition 1, the maximum potential difference in regional average treatment effects attributable to effect modifier distributional differences is

$$\Delta_{\max} = L_{\text{clinical}} \cdot W_1(F_1, F_2), \tag{8}$$

expressed in the units of the trial outcome scale. The formula is invariant to within-effect-modifier scale normalization of W_1: any normalized form W_1/X multiplied by the same scale X cancels in equation (8), so adopting W_1 in its original units yields the same clinical-scale calibration as any rescaled version. When L_{clinical} is independently known, equation (8) provides a direct clinical-scale translation of the observed distributional difference. When L_{clinical} is unknown, the relation can be inverted to ask which L_{clinical} would be required for a given W_1 to produce a clinically meaningful $\Delta_{\max}$.

The estimator $\widehat{W}_1$ has a percentile bootstrap CI (Section 2.3), and inferential uncertainty propagates linearly to $\Delta_{\max}$ when L_{clinical} is treated as fixed: $[\Delta_{\max,L}, \Delta_{\max,U}] = L_{\text{clinical}} \cdot [W_{1,L}, W_{1,U}]$. When L_{clinical} itself is uncertain or unavailable, we solve equation (8) for L_{clinical} at a pre-specified clinical margin Δ_{clin} (the minimal clinically important treatment effect difference, the non-inferiority margin, or the targeted overall effect), yielding the required clinical slope

$$L^* = \frac{\Delta_{\text{clin}}}{W_1(F_1, F_2)}. \tag{9}$$

L^* is the smallest clinical slope under which the observed distributional difference could produce a clinically meaningful regional treatment effect difference. Because L^* is a decreasing function of W_1, the bootstrap CI of $\widehat{W}_1$ gives the percentile interval $[\Delta_{\text{clin}}/W_{1,U},\ \Delta_{\text{clin}}/W_{1,L}]$ for L^*. If a plausible upper bound on L_{clinical} from prior evidence (denoted L_{UB}) lies below L^*, the distributional difference cannot produce such a difference, provided $L_{\text{clinical}} \leq L_{\text{UB}}$; reading L_{UB} against the lower limit $\Delta_{\text{clin}}/W_{1,U}$ rather than the point value carries the estimation uncertainty of $\widehat{W}_1$ into this comparison. The two pathways—forward calibration via equation (8) when L_{clinical} is known and reverse comparison via equation (9) against L_{UB} when L_{clinical} is unknown—cover the two evidence states encountered at the MRCT planning stage.

Specifying L_{clinical} (or its upper bound L_{UB}) is a clinical judgment, informed but not determined by prior evidence, for which the sponsor carries responsibility—the same footing as a non-inferiority margin. Writing $R_a(x)$ for the risk under arm a, $\tau(x) = R_1(x) - R_0(x)$, so $|\tau'(x)| \leq |R_1'(x)| + |R_0'(x)|$: arm-level risk gradients constrain the modification slope only jointly, and a prognostic gradient alone guarantees no bound, prognostic and treatment-effect gradients being distinct quantities.[18,19] For the GUSTO-I case study presented in Section 4, the Fibrinolytic Therapy Trialists' meta-analysis[20], GUSTO-I subgroup analyses,[21] and the GUSTO-I prognostic model[22] inform illustrative bounds of $L_{\text{UB,age}} \approx 1 \times 10^{-2}$ per year and $L_{\text{UB,SBP}} \approx 2 \times 10^{-3}$ per mmHg on the 30-day mortality scale; these exceed the empirical prognostic age gradient of approximately 6×10^{-3} per year derived from the age-stratified mortality reported by Lee et al.,[22] though not the sum of two arm-level gradients of that size that the inequality above permits—a reference point rather than a guarantee. Conclusions drawn from the bounds are conditional on them; because L^* is reported for every region pair, any alternative bound can be compared against it directly. Alternative specification strategies include the deft estimator of Fisher et al.[17]

The pooling assessment proceeds by computing per-effect-modifier W_1 with bootstrap CIs across candidate partner regions, ordering the candidates by the point estimate $\widehat{W}_1$ within each effect modifier, reading each distance against its null floor and its upper confidence limit (Section 2.3)—distances unresolved against the null floor are not ordered among themselves—and deriving $\Delta_{\max}$ when L_{clinical} is known or L^* at a pre-specified Δ_{clin} when L_{clinical} is unknown. The framework is deliberately estimation-centered rather than testing-based: W_1 has no universal cutoff that separates "similar" from "dissimilar," and what a given W_1 means clinically depends on the clinical slope L_{clinical}, which is external to the distributional comparison. By presenting

per-pair $\Delta_{\max}$ (or L^*) with bootstrap CIs alongside the clinical context—the overall treatment effect size, the non-inferiority margin, the minimal clinically important difference—the framework provides quantitative inputs for sponsor and regulatory judgment. The judgment is not driven by a significance test.[23]

3 | SIMULATION STUDIES

We conducted two complementary simulation studies. Study 1 evaluates identification: whether each candidate distance measure correctly identifies pooling regions—in the simulation, the regions built from the same population, whose effect modifier distributions are identical—by ranking candidate partners for an anchor region. Study 2 evaluates estimation performance: bias, variability, and coverage probability of bootstrap confidence intervals for $\widehat{W}_1$. Study 1 carries the primary comparison, because a distance that cannot distinguish differing distributions cannot usefully be calibrated; Study 2 underwrites the calibration pathway that converts an estimated distance into a $\Delta_{\max}$ statement via equation (8).

Because the two studies answer different questions, they produce different sample-size statements. Study 2's $n \geq 100$ per region is an estimation criterion, based on bias and coverage. Study 1's required sample sizes are discrimination criteria: the sample size at which a measure's discrimination reaches an area under the receiver operating characteristic curve (AUC) of at least 0.9, which for $\widehat{W}_1$ ranges from 42 to 690 per region across the cells reported below. The two statements are labelled throughout and are not interchangeable.

3.1 | Study 1: Identification

3.1.1 | Design

Study 1 follows the ADEMP structure (aims, data-generating mechanisms, estimands, methods, performance measures) for simulation studies.[24] Four worlds are simulated, each built from a single distributional family and each containing regions constructed either from the anchor's own population (true matches) or from a discordant population. Set 1 (Gaussian) places discordance in location and scale. Set 2 (log-normal) places it in location and in dispersion combined with shape. Set 3 (mixture) is the adversarial world for summary-based measures: every region, matched or discordant, is constructed with mean 50 and SD 10, so all discordance lies in distributional shape. Set 4 (extremes) places a 5% subgroup in each tail, 40 units from the centre in the anchor; discordant regions move both subgroups to 70 or 100 units or raise their prevalence to 12% (mean fixed), or move only the upper subgroup to 100 units, the pattern of a small severe subpopulation; two control regions instead shift their whole population by 2 and 3 units, a configuration deliberately favourable to CDF-supremum statistics. Ground truth is the construction label—which regions were built from the anchor's population—so the target is defined without reference to any of the compared measures and the comparison is not circular.

Three measures enter the comparison: W_1; the KS statistic, the most common comparison of full distributions; and SMD, the standard comparison of location. On the log-normal world, SMD computed on the log scale is also run; the transformation recovers the shape contrast (AUC 1.00 at n = 2,000) but leaves the combined cell at chance (0.51), so a transformation that suits one contrast does not repair another.

The measures are scored against the construction label on a selection task. An anchor and nine candidate regions are drawn; each measure ranks the candidates by their distance to the anchor and retains the k closest, where k is the number of true matches. Giving every measure the same k removes threshold choice from the comparison entirely; supplying k is itself an oracle, and the application chooses no k but reads the ordering with the clinical calibration (Section 4). Performance is summarized by the AUC for discriminating true matches from discordant candidates (chance = 0.5) and by the probability that the k retained regions include at least one discordant region. The task ran 10,000 replications per cell at the base grid (n = 25–100 per region) and 3,000 at the extended grid (n = 150–2,000). The extension to n = 2,000 is what makes the classification below possible.

3.1.2 | Two Behaviours: Blind and Recovering

Extending the sample-size grid to n = 2,000 separates the measures into two behaviours (Figure 2). A measure is blind in a cell when its AUC stays at chance through n = 2,000: the population value of its distance carries no information about the construction

label, so the task is unsolvable for that measure at any sample size. This is an identification failure, not a power failure. In Set 3 it is exact by construction: all regions there share mean and SD, so the population SMD is zero. On the selection task, SMD is blind in seven of the thirteen cells, counting each world's combined cell: every cell of Set 3, the two mean-matched cells of Set 4, the scale cell of Set 1, and the shape cell of Set 2 (AUC 0.49–0.51 across the full grid in all seven). Adding the SD as a further summary would not help in Set 3, where every region shares mean and SD, nor would adding skewness in the shape-symmetric cell, where it is matched as well: any finite set of summaries can be matched by distributions that differ elsewhere.

A measure is recovering when its AUC rises monotonically through the target: identification succeeds, and what remains is a required-sample-size question. W_1 and the KS statistic are recovering in every cell. In particular, the KS statistic is blind nowhere: in the Set 4 cell where it is weakest it still reaches selection AUC 0.99 at n = 2,000. Its failures in what follows are failures of efficiency, not of identification.

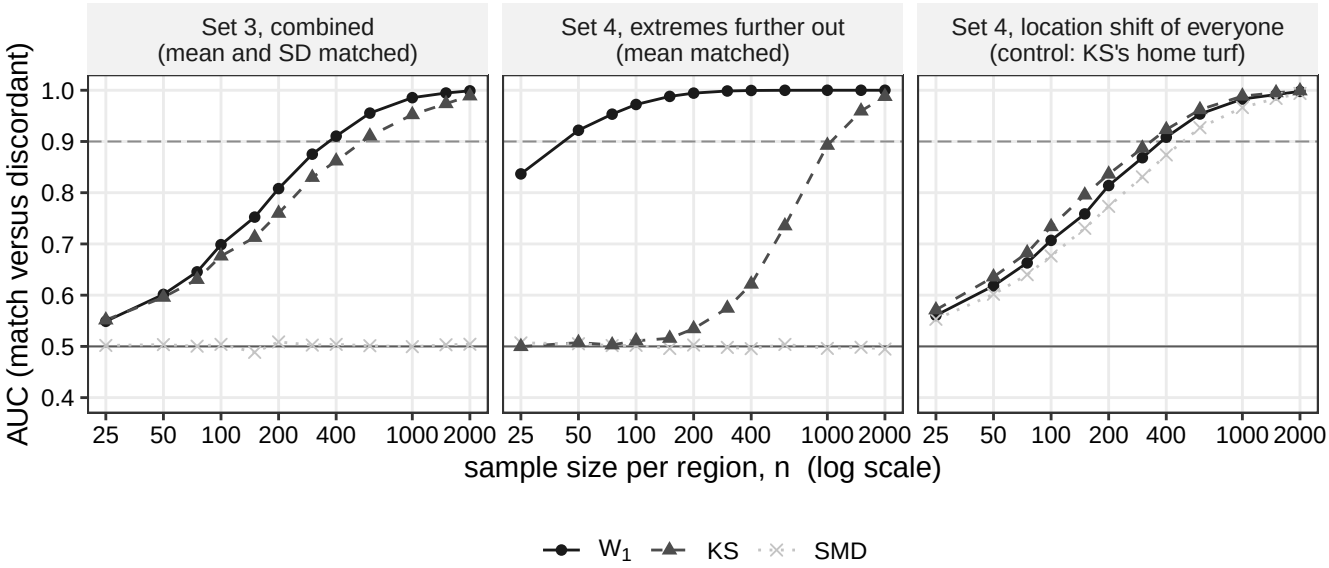


FIGURE 2 Selection AUC (true matches versus discordant candidates) as a function of per-region sample size (n = 25–2,000, log scale) for three distance measures in three cells of Study 1: the Set 3 combined cell (mean- and SD-matched mixtures), the Set 4 symmetric-severity cell (mean-matched displaced extremes), and the Set 4 bulk-shift control cell (uniform location shift of the whole population). The dashed horizontal line marks AUC = 0.9; chance corresponds to 0.5.

3.1.3 | Selection: Required Sample Sizes and Error Rates

Table 1 reports, for every cell of Sets 3 and 4, the worlds built to challenge summary-based measures, the per-region sample size at which each measure's selection AUC reaches 0.9. Two kinds of number appear in it, and they answer different questions. Reading down the W_1 column shows how W_1's requirement varies with the task, from 42 (symmetric severity) to 690 (shape-symmetric mixtures). Reading across a row gives the method difference within one task: in the symmetric-severity cell, W_1 reaches the target at n = 42 and the KS statistic at n = 1,047, a 25-fold difference on the same data-generating process. The control cell behaves as designed: a uniform shift of the whole population is the configuration the CDF supremum is best suited to, the KS statistic wins it (333 versus 377), and we report the loss.

The required-n comparison is complemented by the error rate at a fixed sample size. At n = 100 in Set 4 (all contrast types pooled), the probability that the three retained regions include at least one discordant region is 0.718 for W_1 and 0.971 for the KS statistic (chance 0.988); the proportion of retained regions that are discordant is 0.30 and 0.58 (chance 0.67).

One further difference is a property of population values rather than of sample size. In Set 4, the true anchor distances of the six discordant regions are, on the W_1 scale, 3.0 and 6.0 (symmetric severity), 4.48 (symmetric prevalence), 3.0 (asymmetric

TABLE 1 Per-region sample size at which selection AUC reaches 0.9, by cell and measure, interpolated on $\log n$ over the grid n = 25–2,000. Cells whose AUC stays at chance (0.49–0.51) across the full grid are marked "blind".

Cell	W_1	KS	SMD
Set 4, symmetric severity	42	1,047	blind
Set 4, symmetric prevalence	98	367	blind
Set 4, asymmetric severity	67	1,252	530
Set 4, bulk shift (control)	377	333	489
Set 3, combined	368	550	blind
Set 3, shape-skewed	537	905	blind
Set 3, shape-symmetric	690	716	blind

Cells are contrast types within each world; "combined" pools a world's contrast types.

severity), and 2.0 and 3.0 (bulk shift); on the KS scale, 0.047, 0.050, 0.067, 0.050, 0.072, and 0.107. The orderings disagree: the two bulk-shift regions, at 2.0 and 3.0 on the W_1 scale, are the two most distant on the KS scale, while the region largest on the W_1 scale (6.0) is separated from the region smallest on the KS scale by a KS gap of 0.003. Consequently, for any threshold c between 3 and 6, no KS threshold retains exactly the regions with true $W_1 \leq c$—at any sample size, however the threshold is chosen. This is a statement about orderings on the W_1 scale, which Proposition 1 ties to the Lipschitz class; it is not independent evidence of clinical superiority.

3.2 | Study 2: Estimation Properties

Study 1 compared the measures' ability to discriminate; Study 2 characterizes the estimator on which the calibration relies: bias, variability, and coverage probability of bootstrap confidence intervals for $\widehat{W}_1$. These properties underwrite the clinical calibration framework, as reliable estimation of W_1 directly determines the quality of $\Delta_{\max}$ estimates via equation (8). We report operating characteristics in the original units of the effect modifier, with each scenario specified at $\sigma = 10$; bias and CI width are therefore on the W_1 scale, while coverage is unitless and directly comparable across scenarios.

3.2.1 | Design

We designed seven systematic scenarios for methodological validation, examining controlled distributional differences in location, scale, scale combined with skewness (S6, a mean-matched log-normal), and a location–scale combination. Table 2 summarizes these scenarios with their true W_1 values, in the original units of the effect modifier.

TABLE 2 Study 2 simulation scenarios with true W_1 values.

ID	Description	Distribution 1	Distribution 2	True W_1
S1	Null (identical)	$N(50, 10^2)$	$N(50, 10^2)$	0.00
S2	Location 0.2σ	$N(50, 10^2)$	$N(52, 10^2)$	2.00
S3	Location 0.5σ	$N(50, 10^2)$	$N(55, 10^2)$	5.00
S4	Location 1.0σ	$N(50, 10^2)$	$N(60, 10^2)$	10.00
S5	Scale 1.5×	$N(50, 10^2)$	$N(50, 15^2)$	3.99
S6	Skew (Log-normal)	$N(50, 10^2)$	LogN$(3.787, 0.5^2)$	12.16
S7	Location + Scale	$N(50, 10^2)$	$N(55, 15^2)$	5.83

The Log-normal in S6 has $\sigma_{\log} = 0.5$ and $\mu_{\log} = \log 50 - \sigma_{\log}^2/2 = 3.787$, giving mean 50, CV = 53.3% and skewness 1.75. S7 combines location and scale differences. True W_1 values for S1–S4 are exact ($W_1 = |\mu_1 - \mu_2|$ for equal-variance normals); S5 uses the closed form $W_1 = \sqrt{2/\pi}\,|\sigma_1 - \sigma_2|$; S6 and S7 are obtained by numerical quadrature of $W_1 = \int |F_1(t) - F_2(t)|\,\mathrm{d}t$, with the S6 integral split at $t = 0$ to accommodate the log-normal support.

For each scenario, we generated samples of size n = 50, 100, and 200 per region, reflecting sample sizes typical in MRCT regional subgroups. We performed 10,000 replications per scenario–sample size combination to ensure stable estimates of operating characteristics, with Monte Carlo standard errors below 0.005 for all reported proportions. Bootstrap confidence intervals were computed using B = 2,000 resamples. All simulations were conducted in R version 4.5.1.

Consistent with the estimation-centered framework, we evaluated the operating characteristics of the sample Wasserstein-1 estimator $\widehat{W}_1$ for the population value W_1, reported in the original units of the effect modifier:

1. Bias: $\text{Mean}(\widehat{W}_1) - W_1$
2. Root mean squared error (RMSE): $\sqrt{\text{Mean}[(\widehat{W}_1 - W_1)^2]}$
3. Coverage probability: Proportion of 95% bootstrap CIs containing the true value
4. CI width: Mean width of bootstrap CIs, reflecting estimation precision

3.2.2 | Results

Table 3 presents the bias of $\widehat{W}_1$ across scenarios and sample sizes, in the original units of the effect modifier (scenarios are specified with $\sigma = 10$). The estimator showed positive bias under the null hypothesis (S1), with bias decreasing from 2.46 at $n = 50$ to 1.26 at $n = 200$. This positive bias arises because the non-negativity of W_1 yields $\widehat{W}_1 > 0$ with probability 1 in finite samples, even when the true value equals zero ($F_1 = F_2$ almost everywhere). The estimator remains consistent ($\widehat{W}_1 \to 0$ as $n \to \infty$ under the null), but the convergence is slow near the parameter space boundary.

TABLE 3 Bias of $\widehat{W}_1$ by scenario and sample size, in the original units of the effect modifier.

Scenario	True W_1	$n = 50$	$n = 100$	$n = 200$
S1 (Null)	0.00	2.463	1.759	1.260
S2 (0.2σ)	2.00	0.989	0.472	0.174
S3 (0.5σ)	5.00	0.175	0.051	0.021
S4 (1.0σ)	10.00	0.034	−0.013	0.005
S5 (Scale)	3.99	0.830	0.393	0.187
S6 (Skew)	12.16	0.377	0.203	0.111
S7 (Loc+Scale)	5.83	0.638	0.340	0.193

Bias is defined as $\text{Mean}(\widehat{W}_1) - W_1$, in the units of the effect modifier (scenarios specified with $\sigma = 10$). Results based on 10,000 replications with $B = 2{,}000$ bootstrap resamples.

For the non-null scenarios away from the boundary (S3–S7), bias was small relative to each scenario's true W_1 and diminished with sample size; at $n = 200$ it was at most 0.193 (S7), a small fraction of the corresponding W_1 (Figure 3(A)). For S4 (1.0σ location shift, $W_1 = 10$), bias was negligible (0.034 at $n = 50$, 0.005 at $n = 200$), confirming accurate estimation even for large distributional differences. For scenarios near the boundary (S1, S2), positive bias was more pronounced and diminished slowly with sample size—a well-known phenomenon for non-negative statistics where sampling variability inflates estimates when the true value is close to zero.

Table 4 presents coverage probabilities of the 95% bootstrap confidence intervals. For the non-null scenarios, coverage was near-nominal at $n \geq 100$: S3 (0.948–0.951), S4 (0.945–0.948), S6 (0.942–0.947), and S7 (0.935–0.943). S4 (1.0σ location shift) showed stable coverage across all sample sizes, with no degradation at larger n, confirming satisfactory bootstrap performance even for large distributional differences. S5 (scale) showed moderate undercoverage at $n = 50$ (0.862) that improved to near-nominal at $n = 200$ (0.946). S2 (small location shift) showed undercoverage at $n = 50$ (0.703) due to its near-boundary true value, improving to 0.947 at $n = 200$.

Coverage for S1 (null) is shown as zero in Figure 3(B) because its true W_1 lies at the parameter space boundary. This does not indicate estimator failure—the estimator is consistent—but a percentile interval cannot contain a true value of zero at any sample size, because with continuous data every bootstrap replicate of $\widehat{W}_1$ is positive, and at $F_1 = F_2$ the naive bootstrap is not consistent (Section 2.3); whether an observed distance is distinguishable from no difference is instead read against the null floor. Where coverage falls short in the non-null scenarios (S2, S5 and S7 at smaller n), the interval misses from below: the true value exceeded the upper limit in at most 1.1% of replications in S2, S5 and S7 at any sample size (never in S2), against 2.5% nominal, and the shortfall came from intervals lying above the true value. The upper confidence limit therefore remains a conservative reading of how large a difference could be.

We also evaluated BCa (bias-corrected and accelerated) confidence intervals in preliminary experiments. BCa intervals showed consistently lower coverage than percentile intervals across all scenarios, particularly for near-boundary scenarios. The BCa

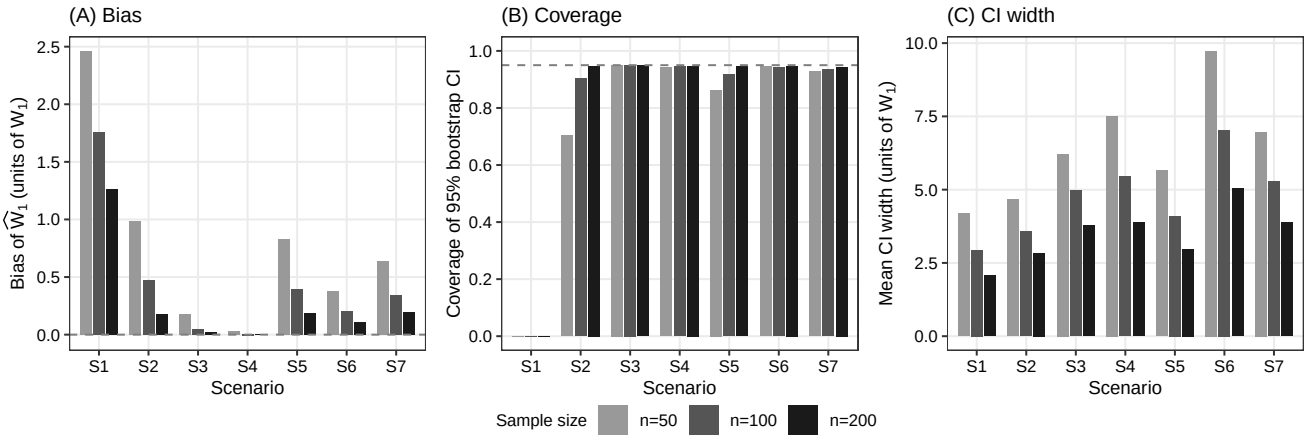


FIGURE 3 Estimation properties of the sample Wasserstein-1 estimator $\widehat{W}_1$, in the original units of the effect modifier, across scenarios (S1–S7) and sample sizes ($n = 50, 100, 200$). Panel (A): bias, with dashed horizontal line at zero. Panel (B): coverage probability of 95% percentile bootstrap CIs, with dashed line at the nominal 0.95 level; S1 coverage is structurally zero because the true value lies at the parameter space boundary (see Section 3.2.2). Panel (C): mean CI width in units of W_1.

correction overcorrects for $\widehat{W}_1$ because the statistic is bounded below by zero, causing the acceleration parameter to distort the quantile adjustment. Based on these findings, we adopt percentile bootstrap as the primary inference method.

Monte Carlo standard errors for all reported coverage probabilities were below 0.005 at 10,000 replications, ensuring that observed differences across scenarios and sample sizes reflect genuine operating characteristic differences rather than simulation noise.

TABLE 4 Coverage probability of 95% percentile bootstrap confidence intervals.

Scenario	$n = 50$	$n = 100$	$n = 200$
S2 (0.2σ)	0.703	0.904	0.947
S3 (0.5σ)	0.951	0.951	0.948
S4 (1.0σ)	0.942	0.945	0.948
S5 (Scale)	0.862	0.917	0.946
S6 (Skew)	0.945	0.942	0.947
S7 (Loc+Scale)	0.928	0.935	0.943

Coverage under S1 (Null) is shown as zero in Figure 3(B) but omitted from this table for brevity; the structural zero arises from the non-negativity constraint at the parameter space boundary.

For S4, coverage was near-nominal across all sample sizes (0.942–0.948), confirming satisfactory performance even for large distributional differences. The undercoverage for S2 at small sample sizes (0.703 at $n = 50$) and S5 (0.862 at $n = 50$) reflects boundary proximity effects that diminish with increasing n.

The mean CI width pattern across scenarios and sample sizes is shown in Figure 3(C). Table 5 presents the RMSE and mean CI width across scenarios, in the original units of the effect modifier. RMSE decreased consistently with increasing n, at the $O(n^{-1/2})$ rate expected from Wasserstein plug-in theory; for S3, for example, it fell from 1.83 at $n = 50$ to 0.98 at $n = 200$. Mean CI width narrowed with n, with CIs at $n = 100$ for the non-null scenarios spanning roughly 3.6–7.0 in the units of the effect modifier. In the clinical calibration framework, this CI width propagates directly to $\Delta_{\max}$ via $\Delta_{\max} = L_{\text{clinical}} \cdot W_1$ (Section 2.4). For example, reading S3 on the age scale with the illustrative bound $L_{\text{UB,age}} = 0.01$/yr of the GUSTO-I application, a $\widehat{W}_1$ CI width of 4.96 years at $n = 100$ corresponds to a $\Delta_{\max}$ CI width of $0.01 \times 4.96 = 0.050$ (5.0%pt), five times the 1%pt margin used in Section 4.4: the estimation criterion does not by itself make the $\Delta_{\max}$ interval narrow relative to a clinical margin.

Under the null scenario (S1), the positive bias produces CIs that are shifted upward from zero (mean CI width 4.19 at $n = 50$ and 2.07 at $n = 200$), so when the true distributional difference is zero the estimator suggests a small positive $\Delta_{\max}$, most at smaller sample sizes. This boundary behaviour is the null floor of Section 2.3 in its simplest instance. Its informative summary is

TABLE 5 RMSE and mean CI width of $\widehat{W}_1$ by scenario and sample size, in the original units of the effect modifier.

	RMSE			Mean CI Width		
Scenario	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
S1 (Null)	2.63	1.87	1.34	4.19	2.94	2.07
S2 (0.2σ)	1.60	1.13	0.85	4.67	3.57	2.84
S3 (0.5σ)	1.83	1.37	0.98	6.22	4.96	3.77
S4 (1.0σ)	1.98	1.42	1.00	7.52	5.45	3.89
S5 (Scale)	1.64	1.11	0.77	5.65	4.10	2.97
S6 (Skew)	2.61	1.86	1.29	9.73	7.04	5.03
S7 (Loc+Scale)	2.00	1.45	1.01	6.97	5.27	3.87

CI width is defined as mean of (upper − lower) across 10,000 replications, in the units of the effect modifier (scenarios specified with $\sigma = 10$).

the upper tail rather than the mean: the 95th percentile of $\widehat{W}_1$ under S1 is 4.21, 2.97 and 2.11 at $n = 50$, 100 and 200 for this scenario ($\sigma = 10$, equal n), and an observed distance at or below the corresponding value is not evidence of a difference.

In summary, the simulation study across seven scenarios demonstrates that the $\widehat{W}_1$ estimator provides reliable estimation with the following properties at sample sizes typical in MRCT regional subgroups:

1. Bias: Small relative to each scenario's W_1 at $n \geq 100$, with positive bias largest near the boundary (S1: 1.76 at $n = 100$) and negligible for the larger-shift scenarios (S3: 0.051, S4: −0.013 at $n = 100$).
2. Coverage: Near-nominal (0.92–0.95) at $n \geq 100$ for the non-null scenarios (S3–S7), including S6 (skew) and S7 (location + scale).
3. Precision: RMSE and CI width decrease at the $O(n^{-1/2})$ rate; at $n = 200$, RMSE ranged from 0.77 (S5) to 1.34 (S1) in units of W_1.

These estimation properties carry over to $\Delta_{\max}$ for a fixed L_{clinical} (Section 2.4), with interval widths that shrink at the rate shown above. For S3–S7, bias was small and coverage near-nominal from $n = 100$ per region; this is the estimation criterion distinguished at the opening of Section 3, and it does not make the $\Delta_{\max}$ interval narrow relative to a given margin. The sample sizes that identification requires are Study 1's (Section 3.1.3).

4 | APPLICATION: HYPOTHETICAL THROMBOLYTIC MRCT USING GUSTO-I DATA

We illustrate the per-effect-modifier Wasserstein-1 framework through a hypothetical scenario grounded in publicly available data from GUSTO-I, a large international trial in acute myocardial infarction (AMI). A sponsor is developing a novel thrombolytic agent (Drug T) for AMI and is planning a Phase 3 MRCT across multiple regions. At the planning stage, the sponsor uses IPD from GUSTO-I ($N = 40{,}830$; 16 regions)[21] to quantify how the distributions of candidate effect modifiers in other regions differ from those in a designated anchor region, treating the GUSTO-I regions as stand-ins for the regional populations the new trial would enrol. GUSTO-I was not designed as an MRCT and its regions are anonymized.

4.1 | Scenario Design and Effect Modifier Selection

For this illustrative exercise, the sponsor identifies two candidate effect modifiers for Drug T's treatment effect in AMI: age and systolic blood pressure (SBP). Drug T's Phase 2 programme has suggested age and SBP as candidate effect modifiers on the basis of exploratory analyses and biological plausibility, but the precision of these analyses is insufficient for numerical estimation of L_{clinical}. At this planning stage the sponsor therefore has no quantitative prior on the clinical slope L_{clinical} for either candidate effect modifier, and class-level indirect evidence is insufficient for a point estimate; it can at most inform a sponsor-specified upper bound (Section 4.4). The Fibrinolytic Therapy Trialists' (FTT) collaborative meta-analysis of existing fibrinolytic agents[20] concluded that benefit is present irrespective of age and of blood pressure, and reports no per-unit CATE slope on the absolute mortality scale for either, so class transferability would not yield a defensible numerical prior for Drug T. We therefore treat L_{clinical} as unknown a priori for both candidate effect modifiers, and apply the L^* reverse-calculation pathway (equation 9) to both.

The sponsor selects Region 8 as the anchor region—representing, for instance, a regulatory market where local efficacy evidence is required. Region 8 comprises n = 2,916 patients in GUSTO-I, with baseline characteristics summarized in Table 6; the anchor role, not its size, is what the exercise illustrates. Quantifying how each region differs from Region 8 mirrors the input a sponsor would bring to a pooling judgment as described in ICH E17.[3]

TABLE 6 Region 8 baseline characteristics (GUSTO-I).

Candidate effect modifier	Mean	SD	Skewness
Age (years)	60.2	12.1	−0.17
SBP (mmHg)	132.4	22.9	0.20

n = 2,916.

4.2 | Null Floor: Which Distances Are Distinguishable from No Difference?

Before interpreting the distances reported in Section 4.3, we computed the null floor of Section 2.3 for each candidate partner. For each partner we drew two independent resamples, with replacement, from Region 8's (R8) empirical distribution at sizes n_{R8} = 2,916 and n_{partner}, recomputed $\widehat{W}_1$, and repeated this 2,000 times, so that the null floor is obtained separately for each partner, as its dependence on both sample sizes requires. We call an observed distance unresolved when it lies at or below the 95th percentile of its null distribution, a reference point for reading the estimate: the observed distance is then one the estimator would produce with appreciable probability from two samples of identical distributions. Table 7 summarises the result.

TABLE 7 Null floor of $\widehat{W}_1$ across the 15 candidate partners, by effect modifier. Ranges are across partners; the floor depends on each partner's sample size. Unresolved distances are those whose observed $\widehat{W}_1$ falls at or below the 95th percentile of the null distribution.

Candidate effect modifier	Null 95th pct.	Unresolved
Age (years)	0.618–0.900	6 / 15
Systolic blood pressure (mmHg)	1.176–1.691	1 / 15

The two candidate effect modifiers differ in how many distances clear the null floor, and the difference matters for how the subsequent distances should be read. On age, the distances of six of the fifteen partners—R1, R4, R5, R7, R9 and R15—are unresolved: they are ones the estimator would produce with appreciable probability from identical distributions. On systolic blood pressure, by contrast, fourteen of fifteen are resolved, the single exception being R2.

Because the null is drawn from one region's distribution, the choice of which region is not neutral: drawing from the anchor asks what $\widehat{W}_1$ would look like if the partner were distributed as R8 is, which is the null appropriate when the partner's data are to inform the anchor. Repeating the computation with each partner's own distribution as the source leaves every distance on the same side of its 95th percentile, so the finding does not depend on which of the two sources is used.

That $\widehat{W}_1$ has a null floor is a property of any plug-in estimator of a non-negative distance, not a defect of this one. Its practical consequence here is that the small age distances of the six unresolved partners are not evidence of a difference, and how large their differences could be is read from the upper confidence limits in Table 8; near this boundary the percentile interval errs by lying above small true values, not below them, in the Study 2 scenarios (Section 3.2.2), so its upper limit is a conservative reading. The partner samples range from 1,231 to 4,352 observations, well above the $n \geq 100$ of Section 3.2.2, so resolution against the null floor is a separate question from the estimation criterion of Study 2.

4.3 | Distributional Assessment: Per-Effect-Modifier W_1 Across 15 Partner Regions

Study 1 showed that W_1 and the KS statistic both reach selection AUC 0.9 in every simulated world, W_1 with fewer patients in every cell but the bulk-shift control; at the GUSTO-I region sizes both suffice, and W_1 is used because its heterogeneity bound has a per-unit clinical slope as its constant (Section 2.2). We therefore computed $\widehat{W}_1$ between Region 8 and each of the remaining 15 regions for both candidate effect modifiers and ordered the partners by it, expressed in the original measurement units (years for age, mmHg for SBP), with percentile bootstrap 95% confidence intervals (B = 2,000). Table 8 presents the results. Figure 4 displays these results as a forest plot with bootstrap CIs, with partners ordered by ascending $\widehat{W}_1$ within each panel.

TABLE 8 Per-effect-modifier Wasserstein-1 distance $\widehat{W}_1$ in original variable units (years for age, mmHg for SBP) with 95% percentile bootstrap CIs for Region 8 versus each partner region (GUSTO-I), listed by region number, with each partner's rank by $\widehat{W}_1$ within each candidate effect modifier.

Partner	n	$\widehat{W}_{1,\text{age}}$ (yr) [95% CI]	Rank	$\widehat{W}_{1,\text{SBP}}$ (mmHg) [95% CI]	Rank
R1	2188	0.65 [0.51, 1.03]	1–6†	3.61 [2.68, 4.93]	7
R2	2952	2.15 [1.57, 2.76]	14	0.95 [0.76, 1.82]	1†
R3	2030	2.61 [1.96, 3.28]	15	3.35 [2.32, 4.65]	4
R4	2876	0.56 [0.37, 1.16]	1–6†	2.60 [1.82, 3.75]	3
R5	1909	0.39 [0.35, 0.92]	1–6†	3.37 [2.32, 4.59]	6
R6	1585	0.91 [0.63, 1.39]	9	3.37 [2.06, 4.77]	5
R7	3150	0.40 [0.30, 0.95]	1–6†	5.07 [4.03, 6.22]	12
R9	3123	0.59 [0.40, 1.12]	1–6†	6.80 [5.71, 8.01]	15
R10	1717	1.38 [0.79, 2.13]	12	4.00 [2.88, 5.39]	8
R11	2491	1.27 [0.76, 1.91]	11	6.44 [5.25, 7.71]	14
R12	4352	0.86 [0.47, 1.43]	8	6.40 [5.39, 7.46]	13
R13	2297	1.15 [0.76, 1.74]	10	2.40 [1.42, 3.72]	2
R14	3437	0.70 [0.42, 1.29]	7	4.36 [3.41, 5.44]	10
R15	2576	0.61 [0.40, 1.25]	1–6†	4.72 [3.60, 5.94]	11
R16	1231	1.74 [1.16, 2.47]	13	4.28 [3.06, 5.71]	9

Point estimates with 95% percentile bootstrap CIs (B = 2,000, seed = 2026). Regions are listed by region number; Rank orders the partners by $\widehat{W}_1$ within each candidate effect modifier (1 = closest to Region 8). †Distances unresolved against their null floor (Table 7) are not ordered among themselves and share the rank range shown. R6 and R5 differ in $\widehat{W}_{1,\text{SBP}}$ by 0.003 mmHg. Bootstrap CIs quantify estimation uncertainty. Narrow CIs lend greater confidence to individual $\widehat{W}_1$ values, while overlapping CIs indicate partners whose relative position is not sharply resolved by the data.

The two candidate effect modifiers produce different patterns of distributional difference. For age, partner-region $\widehat{W}_{1,\text{age}}$ point estimates span from 0.39 years (R5) at the low end to 2.61 years (R3) at the high end, with most partners below 1.5 years. For SBP, point estimates span from 0.95 mmHg (R2) to 6.80 mmHg (R9). The rankings of partners by the two candidate effect modifiers differ.

Several partners fall in the middle of each ranking where uncertainty intervals overlap substantially. R16's $\widehat{W}_{1,\text{age}}$ (1.74 years) is the 13th-smallest value with 95% CI [1.16, 2.47] years, placing it in a region where its relative position is not sharply determined by the data—the CI spans the range covered by several partners with smaller point estimates. R6's $\widehat{W}_{1,\text{age}}$ (0.91 years) is the 9th-smallest value, while its $\widehat{W}_{1,\text{SBP}}$ of 3.37 mmHg sits in the lower-middle of the SBP distribution with 95% CI [2.06, 4.77] mmHg, similarly spanning a broad region of the SBP ranking. These mid-rank cases illustrate the value of reporting bootstrap CIs alongside point estimates. The ranking itself is accompanied by an uncertainty interval that informs how confidently one partner can be placed above or below another.

A contrast emerges between R2 and R9. R2 has the second-largest $\widehat{W}_{1,\text{age}}$ (2.15 years) but the smallest $\widehat{W}_{1,\text{SBP}}$ (0.95 mmHg), while R9 is among the six smallest $\widehat{W}_{1,\text{age}}$ (0.59 years) but the largest $\widehat{W}_{1,\text{SBP}}$ (6.80 mmHg). Both small distances are unresolved against their null floors (Section 4.2), so the reversal rests on the two large ones. A ranking based on either candidate effect modifier alone would order these two regions oppositely. The reversal shows that the distance on one candidate effect modifier carries little information about the distance on another; Section 5 quantifies this across all region pairs.

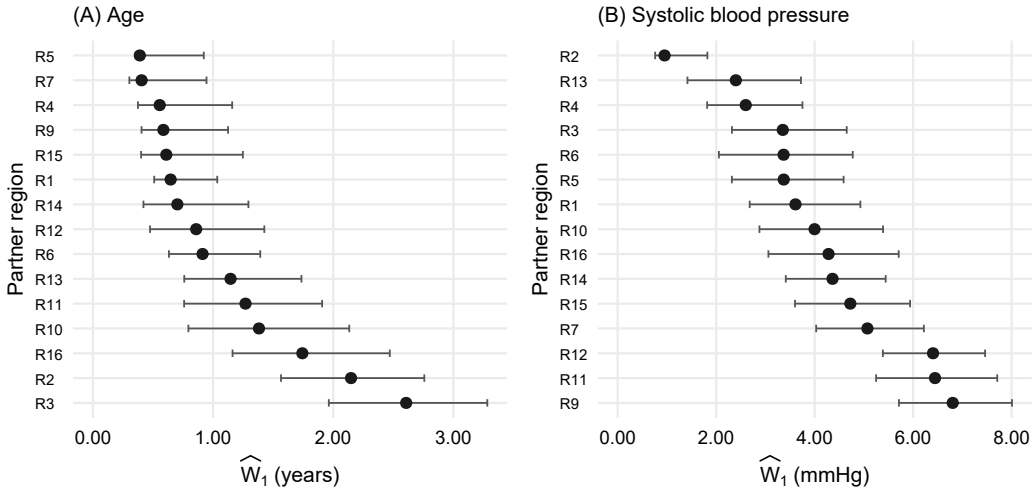


FIGURE 4 Forest plots of $\widehat{W}_1$ point estimates in original units (years for age, panel A; mmHg for SBP, panel B) with 95% percentile bootstrap CIs. Partners are ordered by ascending $\widehat{W}_1$ within each panel; ordering and horizontal scale differ between panels.

4.4 | Clinical Interpretation

Following Section 2.4, we read L^* against illustrative bounds that are the sponsor's clinical inputs, informed by the class-level evidence cited there: $L_{\mathrm{UB,age}} = 1 \times 10^{-2}$ per year (10%pt per decade) and $L_{\mathrm{UB,SBP}} = 2 \times 10^{-3}$ per mmHg (2%pt per 10 mmHg) on the 30-day mortality scale.

Table 9 reports, for each partner region, L^* with its percentile interval (Section 2.4). Each interval is read against the illustrative bound L_{UB} for its candidate effect modifier. Where the whole interval lies above L_{UB}, even the upper confidence limit of the distributional difference would require a clinical slope above the bound to produce $\Delta_{\mathrm{clin}} = 1\%$pt; where the interval contains L_{UB}, the data are compatible with a difference that could and one that could not produce it; where the interval lies below L_{UB}, a slope within the bound could produce it even at the lower confidence limit of the distance. For distances unresolved against the null floor (Section 4.2), only the lower end of the interval, which comes from the upper confidence limit of $\widehat{W}_1$, carries the reading.

5 | DISCUSSION

This paper answers the requirements set out in Section 2.1 and characterizes the estimation of its answer, showing that the W_1 distance: (i) resolves distributional differences to which a location summary such as the SMD is structurally blind—in Study 1, SMD is at chance at every sample size examined in seven of the thirteen selection cells, an identification failure that no sample size repairs, whereas W_1 recovers the true partner structure in all four worlds (Section 3.1.2); (ii) translates distributional distance into a clinically interpretable scale via the heterogeneity bound $\Delta_{\max}$ (equation 8), whose constant—a treatment effect change per unit of the effect modifier—is a clinical slope that the sponsor specifies, informed by clinical evidence, unlike the total-variation constant of the corresponding Kolmogorov–Smirnov bound, and unlike SMD, which supports no bound valid across the Lipschitz class, since regions with equal means can have $W_1 > 0$ and hence, by equation (5), regional treatment effects differing by up to $L_{\mathrm{clinical}} W_1$; and (iii) admits percentile bootstrap intervals with near-nominal coverage for non-negligible differences in the location, scale and skewness scenarios of Study 2 at moderate sample sizes ($n \geq 100$ per region), with a characterized boundary behavior near the null (Section 2.3).

These methodological findings carry two interpretive implications for pooling assessments, illustrated by the GUSTO-I application as a worked example of the framework's operation rather than a substantive epidemiological claim. First, when multiple candidate effect modifiers are under consideration, the per-modifier distances need not agree: a partner close to the anchor on one effect modifier can be distant on another, as exemplified by the contrast between R2 and R9, so the distance on a

TABLE 9 L^* at Δ_{clin} = 1%pt with 95% percentile interval for all 15 partner regions, listed by region number. L^* is the minimum clinical slope (per year for age, per mmHg for SBP) required for the observed distributional difference to produce a treatment effect difference of 1%pt on the absolute 30-day mortality scale.

Partner	$\widehat{W}_{1,\text{age}}$ (yr)	$\widehat{W}_{1,\text{SBP}}$ (mmHg)	L^* age (/yr)	L^* SBP (/mmHg)
R1	0.65†	3.61	0.0155 [0.0097, 0.0196]	0.0028 [0.0020, 0.0037]
R2	2.15	0.95†	0.0047 [0.0036, 0.0064]	0.0105 [0.0055, 0.0131]
R3	2.61	3.35	0.0038 [0.0030, 0.0051]	0.0030 [0.0022, 0.0043]
R4	0.56†	2.60	0.0180 [0.0086, 0.0267]	0.0038 [0.0027, 0.0055]
R5	0.39†	3.37	0.0256 [0.0108, 0.0282]	0.0030 [0.0022, 0.0043]
R6	0.91	3.37	0.0110 [0.0072, 0.0158]	0.0030 [0.0021, 0.0049]
R7	0.40†	5.07	0.0247 [0.0106, 0.0330]	0.0020 [0.0016, 0.0025]
R9	0.59†	6.80	0.0171 [0.0089, 0.0247]	0.0015 [0.0012, 0.0018]
R10	1.38	4.00	0.0072 [0.0047, 0.0126]	0.0025 [0.0019, 0.0035]
R11	1.27	6.44	0.0079 [0.0052, 0.0132]	0.0016 [0.0013, 0.0019]
R12	0.86	6.40	0.0116 [0.0070, 0.0211]	0.0016 [0.0013, 0.0019]
R13	1.15	2.40	0.0087 [0.0058, 0.0132]	0.0042 [0.0027, 0.0071]
R14	0.70	4.36	0.0142 [0.0077, 0.0238]	0.0023 [0.0018, 0.0029]
R15	0.61†	4.72	0.0164 [0.0080, 0.0249]	0.0021 [0.0017, 0.0028]
R16	1.74	4.28	0.0057 [0.0040, 0.0086]	0.0023 [0.0018, 0.0033]

Regions are listed by region number; their ranks by $\widehat{W}_1$ are given in Table 8. †Distance unresolved against its null floor (Table 7). The illustrative bounds of Section 4.4 are $L_{\text{UB,age}} = 1 \times 10^{-2}$ /yr and $L_{\text{UB,SBP}} = 2 \times 10^{-3}$ /mmHg; the lower limit of R1's SBP interval, 0.00203, lies just above $L_{\text{UB,SBP}}$.

modifier left out of the assessment cannot be inferred from those included. Second, W_1 ranking alone does not convey clinical weight. Within one effect modifier, $L^* = \Delta_{\text{clin}}/W_1$ is a strictly decreasing function of W_1, so calibration does not reorder partners relative to their distances; what it adds is the comparison of each partner's L^* with the bound L_{UB} specified for that effect modifier, which places a distance measured in the modifier's own units on the outcome scale. The distance and its calibration therefore carry different information, and both are reported for each candidate effect modifier.

These implications also bear on the cluster analysis of regions that Komiyama et al. introduce as one example of forming pooled regions, in which the candidate effect modifiers are combined into a single weighted distance.[6] The per-effect-modifier resolution adopted here matters precisely because partner rankings can reverse across modifiers: in the GUSTO-I example R2 and R9 each rank similar on one modifier but dissimilar on the other. The reversal is not confined to these two regions: across the 15 partners of Table 8, the orderings by age and by systolic blood pressure distance do not agree (Spearman correlation −0.30), so a single distance aggregated over the two modifiers would mask differences that a per-modifier assessment retains. How many pooled regions to form, and how orderings that differ across modifiers are reconciled, remain clinical judgments.

Section 2.1 set out the limitations of the existing measures: SMD captures only location; the KS statistic and related EDF tests lack a clinically interpretable scale, and the heterogeneity bound they admit is governed by the total variation of the CATE, a constant clinical sources do not report; and KL divergence is asymmetric, can diverge under non-overlapping support, and requires density estimation impractical at small regional samples. The W_1 distance addresses each. The location-only limitation is structural, and Study 1 exhibits its consequence: in the mean-matched cells, SMD is at chance at every sample size examined while W_1 recovers the partner structure (Section 3.1.2). The lack of a clinical scale is addressed by the heterogeneity bound (equation 8), which translates distributional distance into clinically meaningful units. The limitations of KL divergence are avoided by construction: W_1 is symmetric, finite for any pair of empirical distributions, and computable directly from CDFs without density estimation, while remaining amenable to bootstrap inference. Addressing these limitations does not, however, mean returning a different answer in every dataset. In GUSTO-I, whose regional differences are predominantly differences in location (partner standard deviations within a factor of 1.11 of the anchor's), W_1 ordered the 15 partners much as the SMD did (Spearman correlation 0.90 for age and 0.95 for systolic blood pressure), and its contribution there was the clinical calibration rather than a different answer. The limitations bind where distributional differences go beyond location—mixtures and rare displaced subgroups, configurations that cannot be excluded in advance—and which kind of dataset one holds is not knowable before the distances are computed. The reason to use W_1 is therefore not that it always gives a different answer, but that, of the measures examined, it recovers in every world, reaches the selection target at a smaller sample size than the KS statistic in every cell of Sets 3 and 4 except the bulk-shift control (Table 1), and carries a translation to the clinical scale whose constant is the per-unit clinical slope.

Two features of the framework bear on its use in practice. First, the calibration adapts to the evidence available on the clinical slope rather than imposing a fixed W_1 cutoff: at the planning stage, where L_{clinical} is typically unavailable for a novel agent, the L^* reverse-calculation (equation 9) is the primary calibration tool; when the sponsor specifies L_{clinical}, informed by prior evidence, the $\Delta_{\max}$ pathway (equation 8) provides a complementary calibration on the clinical scale. Second, the distributional assessment and its calibration are separable. The identification and efficiency results of the preceding paragraph are properties of the distance itself and hold whether or not L_{clinical} is available; clinical calibration adds an outcome-scale reading but does not change them.

Because W_1, like other covariate-distribution measures, requires only baseline effect modifier distributions, regional data can be assembled from prior trials, registries, electronic health records, or other real-world evidence (RWE) sources, provided the source population approximates the population each region is expected to enrol. The assessment can therefore be made at the planning stage, before any regional data from the new trial exist.

The W_1 distance is one input within a holistic regulatory evaluation; biological plausibility, clinical relevance, and benefit–risk balance extend beyond what any single distributional measure can quantify.[3,4] The framework has further limitations. It treats continuous effect modifiers marginally, leaving categorical and multivariate extensions for future work. Because W_1 is non-negative, $\widehat{W}_1$ exhibits positive bias near zero, zero percentile coverage when the true value is zero (S1), and undercoverage for small distances (S2: 0.703 and S5: 0.862 at $n = 50$), with the intervals erring above the true value (Section 3.2.2);[25] estimates near the null are read against the null floor, and the estimation criterion of Study 2 is $n \geq 100$ per region. Proposition 1 also assumes a CATE common to all regions; regional differences in extrinsic factors that alter τ itself, such as standard of care or concomitant therapy, lie outside what W_1 measures. Finally, as with any covariate-based approach, unmeasured effect modifiers may contribute to heterogeneity beyond what the distance captures.

The calibration rests on the Lipschitz premise of Proposition 1, and its limitations are those of the premise. First, the constant is a clinical judgment: L_{UB} is specified by the sponsor, informed but not determined by prior evidence (Section 2.4), so conclusions drawn from $\Delta_{\max}$, or from the comparison of L^* with L_{UB}, are conditional on it; reporting L^* leaves the comparison open to any alternative bound. Second, an L_{clinical} taken from an estimated average interaction slope is a lower bound on the Lipschitz constant rather than the constant itself; the safety factor of Section 2.4 is a judgment of the same kind as margin discounting, and the framework does not supply its value. Third, for a CATE that changes steeply over a narrow range of the effect modifier—a threshold or step function—the Lipschitz constant is large and equation (6) is correspondingly loose. This is by design: the bound is a worst case over the CATEs compatible with the stated L_{clinical}, and a loose bound overstates rather than understates the potential difference. In that case the Kolmogorov–Smirnov bound of Section 2.2, whose constant is the total excursion of the CATE, may be the tighter of the two, and a sponsor able to state the size of a threshold effect has that alternative. The requirement is also local: since $\bar{\tau}_1 - \bar{\tau}_2 = -\int (F_1 - F_2)\, d\tau$, the bound holds with L_{clinical} replaced by the Lipschitz constant of τ on $\{x : F_1(x) \neq F_2(x)\}$, so saturation of the CATE outside that range does not loosen it; for distributions with full support, such as normal distributions, that range is the whole line. Finally, the bound is per effect modifier: it assumes the CATE varies with that modifier alone. When the CATE is additive across modifiers, $\tau(x) = \sum_k \tau_k(x_k)$, the per-modifier bounds combine as $|\bar{\tau}_1 - \bar{\tau}_2| \leq \sum_k L_k W_1^{(k)}$, and each per-modifier $\Delta_{\max}$ is the contribution attributable to that modifier; under interaction between modifiers they do not, because marginal distances do not control the joint distance.

Beyond the scope and methodological extensions noted above, two further directions warrant investigation. The upstream identification of which baseline characteristics constitute relevant effect modifiers is itself a non-trivial problem deserving dedicated treatment; regression-based relevance weighting such as the Lasso offers one front-end for this selection,[6] after which the present framework quantifies distributional distance on the retained modifiers. Bias-correction methods tailored to the boundary behavior of $\widehat{W}_1$ at small samples could extend the operational range of inference.

The principal contribution of this paper is a distributional similarity measure for effect modifiers whose calibration constant is a per-unit clinical slope. The W_1 distance, combined with bootstrap confidence intervals and clinical calibration via $\Delta_{\max}$ or L^*, supplies a reproducible quantitative basis for the judgment of "similar enough", operationalizing the similarity of effect modifier distributions that ICH E17 describes, and informing, not prescribing, pooling decisions.

AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

R code for computing the Wasserstein-1 distance and reproducing the simulation studies and the application is available at [repository URL]. The GUSTO-I individual patient data analysed in Section 4 are publicly available as the dataset `gusto` in the R package predtools (version 0.0.3; https://github.com/resplab/predtools); the trial is described in the original publication.[21]

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website. Supporting materials include the R code for all analyses.

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APPENDIX

A PROOFS AND MATHEMATICAL DETAILS

A.1 Non-negativity and Identity of Indiscernibles

Non-negativity is immediate: $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx \geq 0$, since the integrand is non-negative.

For the identity of indiscernibles, we show both directions. If $F_1 = F_2$, then $|F_1(x) - F_2(x)| = 0$ for all x, so $W_1(F_1, F_2) = 0$. Conversely, suppose $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx = 0$. Because the integrand is non-negative, it vanishes for Lebesgue-almost every x, so $F_1 = F_2$ almost everywhere. Right-continuity then upgrades this to equality everywhere: a null set contains no interval, so for any x and every $\delta > 0$ the interval $(x, x + \delta)$ contains points at which $F_1 = F_2$; choosing such $x_j \downarrow x$ and letting $j \to \infty$ gives $F_1(x) = F_2(x)$. Hence $F_1 = F_2$.

A.2 Asymptotic Properties

Under standard regularity conditions, the empirical Wasserstein-1 distance $\widehat{W}_1 = W_1(\hat{F}_1, \hat{F}_2)$ is a consistent estimator of $W_1(F_1, F_2)$.[11]

In one dimension, the empirical W_1 distance converges at rate $O(n^{-1/2})$ if and only if $\int \sqrt{F(x)(1 - F(x))}\, dx < \infty$.[11] For two samples with $n_1/(n_1 + n_2) \to \lambda \in (0, 1)$, when $\int \sqrt{F_r(1 - F_r)}\, dx < \infty$ $(r = 1, 2)$ and $\{x : F_1(x) = F_2(x) \in (0, 1)\}$ has Lebesgue measure zero, the directional derivative of W_1 at (F_1, F_2) is linear, which is the condition under which the bootstrap statement below holds;[13] this condition excludes $F_1 = F_2$. At the boundary $F_1 = F_2$ the derivative is non-linear and the limit is not Gaussian; separately, a percentile interval cannot contain zero, which is the degenerate coverage observed under the null (S1; Section 3.2.2). For an effect modifier recorded on a grid, the CDFs are step functions and the condition fails whenever they agree at any grid point inside the support; the finite-space results of Sommerfeld and Munk then apply.[15]

Under the same conditions the percentile bootstrap confidence interval achieves asymptotically correct coverage;[13] where the two CDFs coincide on an interval, including $F_1 = F_2$, the naive bootstrap is not consistent.[14,13]

B R CODE FOR W_1 COMPUTATION

Listing 1 R function for computing the Wasserstein-1 distance with bootstrap confidence intervals.

```
compute_W1 <- function(x1, x2) {
  pooled <- c(x1, x2)
  all_vals <- sort(unique(pooled))
  F1 <- ecdf(x1); F2 <- ecdf(x2)
  w1 <- sum(diff(all_vals) *
    abs(F1(all_vals[-length(all_vals)]) -
        F2(all_vals[-length(all_vals)])))
  w1
}

W1_bootstrap <- function(x1, x2,
    B = 2000, conf = 0.95) {
  obs <- compute_W1(x1, x2)
  boot_vals <- replicate(B, {
    b1 <- sample(x1, replace = TRUE)
    b2 <- sample(x2, replace = TRUE)
    compute_W1(b1, b2)
  })
  alpha <- 1 - conf
  ci <- quantile(boot_vals,
    c(alpha/2, 1 - alpha/2), na.rm = TRUE)
  list(estimate = obs,
       ci_lower = ci[1], ci_upper = ci[2])
}
```